Encrypted USB drives and removable media

Article KB0010102 | Owner: Security Operations | Subcategory: General | Applies to: Windows 11 23H2/24H2 laptops, macOS 14/15 MacBook Pro/Air, Any USB flash drive, external HDD/SSD, SD card used for Acme data

Summary

This article explains how Acme Global employees can securely use encrypted USB drives and other removable media, outlines the self-help steps to encrypt a drive with BitLocker To?Go (Windows) or FileVault (macOS), and provides the service-desk procedure for approving, issuing, and supporting encrypted media.

Symptoms

- USB drive not recognized after insertion
- BitLocker prompts for recovery key when trying to access the drive
- macOS asks for FileVault password and you do not know it
- Error message: 'The disk is not formatted' or 'Access is denied'

Common causes

- Drive was not encrypted with Acme-approved method before use
- Recovery key not backed up to Intune, causing lockout after forgotten PIN/password
- Using a personally owned USB stick that lacks corporate encryption

Overview

Acme Global requires that any removable media used to store or transfer company data be encrypted with an approved solution. For Windows devices this is BitLocker To?Go; for macOS devices it is FileVault. Encryption protects data if the media is lost or stolen and satisfies the Data Protection Policy. Only media that has been provisioned through the IT Service Desk or that you have encrypted yourself following the self-help steps below may be used for Acme data.

Self-Help Procedure: Encrypting a USB Drive (Windows)

Follow these steps to encrypt a USB flash drive or external disk using BitLocker To?Go. The process stores the recovery key in Azure AD/Intune so you can recover it if you forget your PIN or password.

1. Insert the USB drive into a free USB port on your Acme-managed laptop.
2. Open File Explorer, right-click the drive, and select "Turn on BitLocker".
3. Choose "Use a password to unlock the drive" and enter a strong password (minimum 14 characters, meeting Acme's password policy).
4. Click "Next" and select "Save the recovery key to your Microsoft account" (this uploads the key to Intune).
5. Choose encryption mode: "Encrypt used disk space only" (faster) or "Encrypt entire drive" (more secure for new drives).
6. Click "Start encrypting". Wait for the progress bar to complete - this may take several minutes depending on drive size.
7. When encryption finishes, you will see a lock icon on the drive in File Explorer. Eject the drive safely; it is now ready for Acme use.

Self-Help Procedure: Encrypting a USB Drive (macOS)

macOS users can encrypt removable media with FileVault. The recovery key is stored in Intune when the drive is encrypted via the Company Portal app.

1. Connect the USB drive to your Mac.
2. Open Finder, locate the drive, right-click (or Control-click) it, and choose "Encrypt \"Drive Name\"".
3. Enter a password that meets the 14-character complexity rule and verify it.
4. Hint: optionally add a hint that does not reveal the password.
5. Click "Encrypt". macOS will begin encrypting the drive; a progress bar shows the status.
6. When complete, the drive will appear with a lock icon. Eject it safely. The encryption password is required each time the drive is mounted.

Service-Desk Agent Procedure

Agents in the Security Operations team handle requests for approved encrypted media, verify compliance, and assist with recovery key retrieval. Follow the workflow below when a user reports an issue with encrypted removable media or requests a new encrypted drive.

1. Verify the user's identity via Entra ID MFA (push or SMS).
2. Determine whether the user needs a new encrypted drive or assistance with an existing one.
3. If a new drive is needed: check inventory for an Acme-approved, FIPS-validated USB flash drive or external SSD.
4. Use the Intune console to create a BitLocker To?Go policy (Windows) or FileVault profile (macOS) that forces encryption and backs up the recovery key to the device's Azure AD record.
5. Physically label the drive with the asset tag and hand it to the user, providing the self-help encryption steps if they prefer to do it themselves.
6. If the user cannot access the drive: retrieve the recovery key from Intune > Devices > select device > BitLocker recovery keys (Windows) or FileVault recovery key (macOS).
7. Provide the key to the user and advise them to change the password/PIN after regaining access.
8. Document the incident in the ticket, noting drive serial number, encryption method, and any user actions taken.

Verification

After encryption, users should confirm that the drive is properly protected before storing any Acme data.

1. Re-insert the drive and verify that a password/PIN prompt appears.
2. Open the drive and attempt to create a small test file; if successful, encryption is working.
3. Check the BitLocker/FileVault status: on Windows, open Control Panel > BitLocker Drive Encryption and confirm the drive shows "Protected"; on macOS, open Finder > Get Info on the drive and look for "Encrypted" under General.

Best Practices & Policies

Always use Acme-provided or pre-approved encrypted media for company data. Never store Acme information on a personal, unencrypted USB stick. Report lost or stolen encrypted media immediately to the Security Operations team via the IT Help Teams channel or phone extension 4357. Recovery keys are retained in Intune for 90 days; after that period they are purged unless a legal hold applies. Keep your encryption password/PIN confidential and do not share it via email or chat.

When to escalate

If a user cannot recover access to an encrypted drive after exhausting the self-help recovery-key retrieval steps, or if the drive shows signs of tampering, malware, or hardware failure, the L1 agent should escalate to the Security Operations group. Include in the escalation: user name, employee ID, drive serial number, encryption method (BitLocker/FileVault), steps already taken, and any error messages or screenshots.

Keywords: USB encryption, BitLocker To Go, FileVault, removable media, encrypted flash drive, recovery key, Intune, secure removable media, data loss prevention, lost USB